

WOW Digital Humanities Toolkit

User Guide

Overview

The WOW Digital Humanities Toolkit is a tool for the automated analysis of PDF documents, designed for researchers and practitioners in the field of Digital Humanities. The program processes the uploaded document, extracts its text via OCR, and produces an HTML report containing charts, lists of recognized entities, and lexical analysis.

The output file is automatically saved in the same folder as the source PDF, with the same name and the .html extension. Simply open it in any browser to view the results.

Requirements

The program is compatible with Microsoft Windows.

Interface and Usage

The main window contains all the options needed for processing. Below is a description of each field.

PDF

Path to the document to be analyzed. Clicking *Browse* allows you to select the desired file using the file explorer. Once selected, the document name will appear in the text field and will be used as the title of the generated HTML report.

Pages – Page Range

The *Start* and *End* fields allow you to define the range of pages to process.

- The default value of *Start* is 1 (first page of the document).
- *End* can be left empty to process the document up to the last available page.
- Negative values are also accepted: for example, -1 indicates the last page, -2 the second-to-last, and so on.

Options

Names Merging

Enables or disables the automatic merging of person names. When enabled, occurrences of isolated surnames are added to those of the full name containing them, provided the full name appears at least twice in the text or is present in the WOW Knowledge Graph.

Example: "Dario Fo": 1 + "Fo": 1 becomes "Dario Fo": 2. The function supports surnames consisting of one or two words (e.g., "De Niro").

Merged names are highlighted in bold in the *Persone* list of the report.

NER Correction

Enables or disables automatic correction of conflicts in entity recognition. When enabled, if the same term is alternatively recognized as a person or a location, the program assigns it to the category in which it appears more frequently.

Margins

Allows you to exclude peripheral areas of the page from OCR, useful for removing recurring elements such as page numbers, headers, or footnotes. Values are expressed as a proportion of the page dimension (from 0 to 1). If a value outside this range is entered, the parameter is automatically reset to its default value.

Parameter	Default Value	Description
Header	0.10	Top margin (10% of height)
Footer	0.10	Bottom margin (10% of height)
Left	0.05	Left margin (5% of width)
Right	0.05	Right margin (5% of width)

Starting the Processing

After selecting the file and configuring the desired parameters, press the *Process* button and wait for completion. When finished, the *Open HTML* button will activate automatically: clicking it will open the report directly in the system's default browser.

HTML Report Structure

The generated HTML file is a self-contained report that does not require an internet connection to be viewed. It is organized into the following sections.

Statistics – Charts

- Pie Chart – *Persone*: shows the most cited people (up to 8), with an additional *W* category for people present in the WOW Knowledge Graph and an *Altri* category for the remaining occurrences.
- Histogram – *Date*: shows the distribution of years detected in the text.
- Histogram – *Luoghi*: shows the 8 locations by number of occurrences.

Keywords and Collocations

For each WOW project keyword present among the 51 most frequent nouns, a collocation graph is generated: words that appear most often nearby (within a window of 3 lemmas to the left and right). Nouns, verbs, and adjectives with at least two occurrences are included.

Annotated Text

The full text extracted from the PDF, with color highlighting of recognized entities.

Category	Description
PERSONE	Person names detected in the text
LUOGHI	Toponyms and geographical references
ORGANIZZAZIONI	Institutions, companies, organizations
DATE	Years detected via regular expressions

Entity and Lemma Lists

At the bottom of the report are the complete lists of recognized entities (people, places, organizations, dates) and the 51 most frequent nouns in the text. For each entry, the number of occurrences is indicated. Terms that correspond to WOW project keywords are marked with the symbol ^w in superscript.

Main Components

Below are the libraries and modules that make up the tool's processing pipeline.

Library	Function
Tkinter	Builds and manages the program's graphical user interface.
PyMuPDF	Rasterizes the PDF document page by page, producing a double-resolution PNG image (zoom ×2) to optimize OCR quality.
PIL / PILLOW	Crops the image margins based on the set parameters, removing unwanted elements such as page numbers.
Pytesseract / Tesseract	Performs optical character recognition (OCR) on Italian text, correctly handling complex layouts such as multi-column text.
Stanza NLP	Performs tokenization, multi-word token expansion (MWT), POS tagging, lemmatization, and named entity recognition (NER) on the text.
Matplotlib	Generates the pie charts, histograms, and bar charts included in the HTML report.
NetworkX	Builds and visualizes the collocation graphs for the project keywords.

References

- **Tkinter:** <https://docs.python.org/3/library/tkinter.html>
- **PyMuPDF:** <https://pymupdf.readthedocs.io/>
- **Pillow:** <https://pillow.readthedocs.io/>
- **Pytesseract:** <https://github.com/madmaze/pytesseract>
- **Tesseract OCR:** <https://github.com/tesseract-ocr/tesseract>

- **Stanza NLP:** <https://stanfordnlp.github.io/stanza/>
- **Matplotlib:** <https://matplotlib.org/stable/>
- **NetworkX:** <https://networkx.org/documentation/stable/>